A REFORMULATION OF THE P-RING CONDITION IN TERMS OF QUOTIENTS BY PAIRS OF PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. The author's proof consists of exhibiting the identity that identifies the two sides; that identity is the whole mathematical content of the equivalence, so the proof is complete rather than omitted, even though it is only a few lines long.

Allowed standard prerequisites (editorial).

- The definition of a P-ring and the property P being tested on the algebras appearing in the statement. the standing definitions, used at proof lines 12850-12855. Fix the two conditions whose equivalence is asserted.

Source reference. Original lemma 0BIS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14500–14500. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a Noetherian ring. Let P be a property as above. Then R is a P-ring if and only if for every pair of primes $\mathfrak{q} \subset \mathfrak{p} \subset R$ the $\kappa(\mathfrak{q})$-algebra*

$$(R/\mathfrak{q})^\wedge_\mathfrak{p} \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q})$$

has property P.

Proof. This follows from the fact that

$$R^\wedge_\mathfrak{p} \otimes_R \kappa(\mathfrak{q}) = (R/\mathfrak{q})^\wedge_\mathfrak{p} \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q})$$

as algebras over $\kappa(\mathfrak{q})$. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.